

Nemagari-no-Takumi Workshop

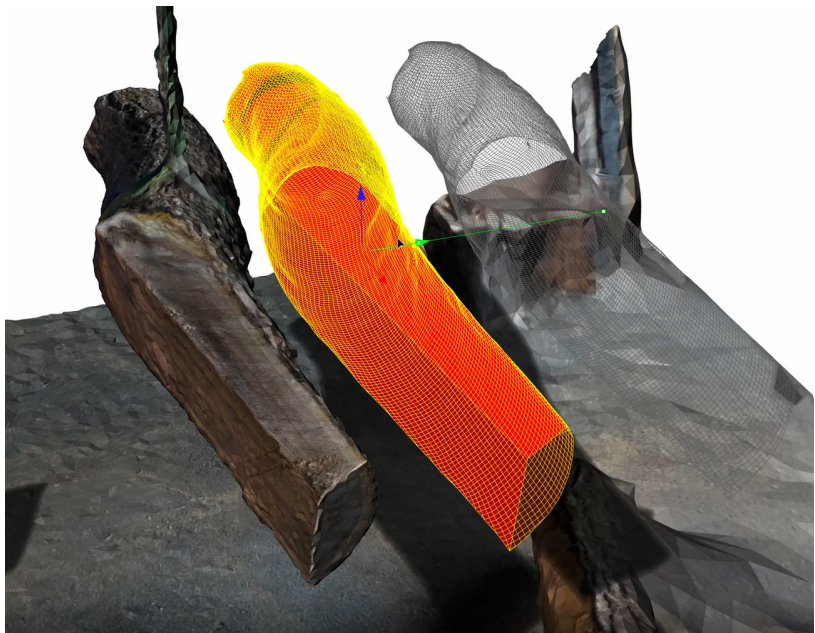
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Mixed-reality Crafting with Unwieldy Logs

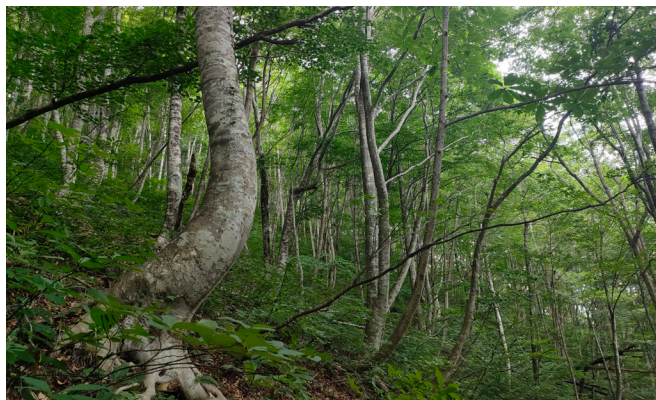


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Following the second world war, Japan's Ministry of Agriculture and Forestry began an aggressive planting campaign to address the country's reconstruction. To enhance the productivity of national forests, huge amounts of cedars were planted due to their quick and resilient growth, replacing the native broad-leaf species in many areas. Simultaneously, Japan's cultural connection to cedar as a building material waned as the country continued to westernize, and eased regulations on imported lumber reduced the country's self-sufficiency rate to 18% by the turn of the century (Sugimoto 2007).

Over time, vast amounts of abandoned tree plantations have grown too dense, blocking sunlight for smaller plants that contribute to healthy biodiversity. This has resulted in increased risk of landslides, increased pollen, and changes to the ecosystem as excessive nitrogen in the soil due to human activity washes into rivers and ponds causing algal blooms as the smaller plants are not present to absorb it (Ru and Chiwa 2021). The forests surrounding Hida, a town in Gifu, Japan are facing these challenges. This region is known historically as a center for Japanese carpentry and is home to a wide variety of traditional crafts that remain in practice today. These mountains are also steep and receive heavy snow, often bending younger trees at their trunk under the load. The tree will continue to grow with a curve, making its lumber difficult to harvest and process within standard industrialized processes. Adding to the region's legacy, when rice was the tax, and since Hida is too mountainous to grow enough to contribute, local carpenters were instead commissioned to build temples and shrines throughout Japan (Imori 1912).

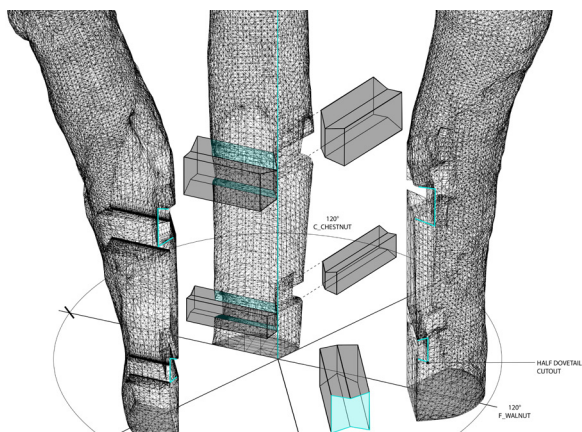
1 Mixed-reality chainsaw cutting
and 3d scan verification



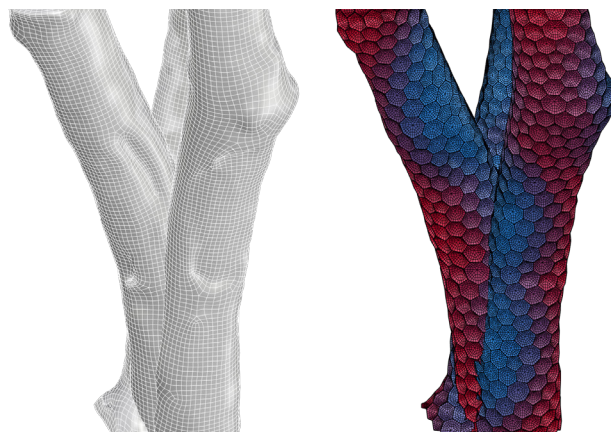
2 Nemagari in a forest near Hida, Japan (Hidakuma 2022)



3 3D scan data of nemagari trees, Hida, Japan (Hidakuma 2022)



4 Refined meshes from 3D scans prepared for MR-guided chainsaw cutting



5 Refined, textured meshes from 3D scans prepared for MR-guided sculpting

"Bent-Root" Timber

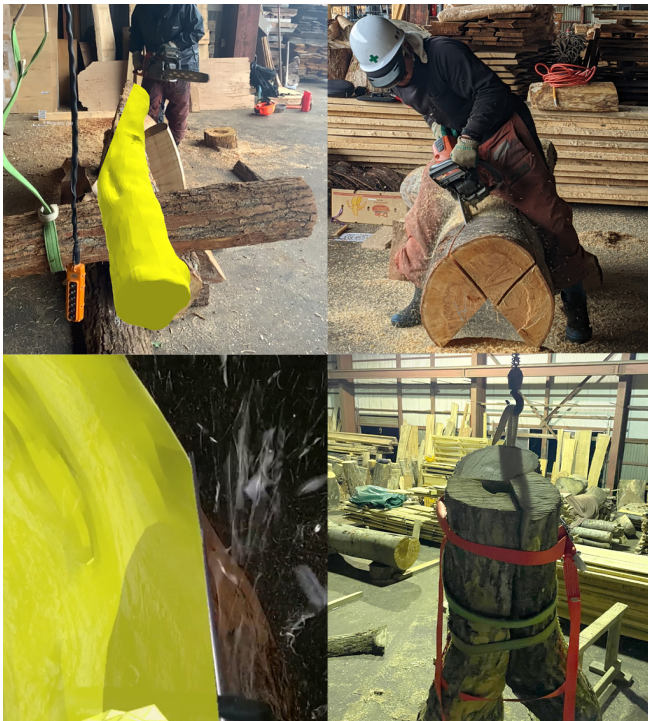
Within this context, this project reflects a recent collaborative and international workshop that connected faculty and students based in the USA with craftspeople and consultants that reside and work in Hida, Japan. The team developed mixed-reality (MR) fabrication workflows to utilize large-scale *nemagari* (bent-root) timber. Nemagari is typically avoided, but when harvested out of necessity, they are reduced to chips or firewood due to their shape. The material remains underutilized as the trees' natural geometry is considered a downgrade in standardized milling operations (Amtsberg et al. 2020).

The MR workflow is intended to enhance the skill of the craftsperson with real-time holographic guides, enabling efficient processing of large and unwieldy logs into architectural elements that make use of their natural curvature which may otherwise be a hindrance to manual, CNC, and robotic fabrication techniques. Construction from an inventory where no two parts are exactly alike necessitates new tools for visualizing the design intent and for accurately cataloging the exact shape of the logs (Lok and Bae 2022). For example, LiDAR scanning was used to verify accuracy prior to and following each chainsaw cut, so that the holographic guides may be adjusted as necessary.

Nemagari Crafting Workshop

The workshop attempted to bridge technological epistemologies, manual and digital, and seamlessly connect faraway, interdisciplinary collaborators. Results shown are a work in progress, pointing toward numerous ways to fine-tune the process and toward other novel uses for MR crafting. The primary toolset consisted of digital modeling and simulation in Rhino/Grasshopper, mixed reality with Microsoft HoloLens 2, and Fologram. Team efforts were divided into four trajectories, each with specific visualization and calibration criteria: structure, balance/weight distribution, texturing, and weaving/thatching.

The "structure" project was the focus of the three-day workshop that resulted in a branching assembly of large nemagari, each approximately 3 meters in length. The 3D scan data for each log was combined and modified into a three-part assembly, requiring two long chainsaw cuts guided by intersecting holographic planes or a full holographic twin of the desired part calibrated to each log as per the chainsaw operator's request (Figure 1). A 3d scan of each cut log was compared to the digital model and necessary adjustments were made to ensure the cut logs align when assembled. Oversized butterfly joints were intended to hold the logs



6 MR chainsaw cutting and assembly



7 Interactive holograms showing a collaborative MR-guided sculpting process

together into a column-like assembly, the joinery of which would be done using hand tools also guided by calibrated holograms. Future work includes tool calibration to track the exact location of the chainsaw blade providing real-time feedback of the blade's position relative to a target cut surface or desired cut depth, and the implementation of MR chainsaw cutting and hand tooling in a completed structure.

With an appreciation for the cultural relationship to trees exemplified in various *matsuri* (civil or religious festivals), the "balance" project focused primarily on ways in which holographic visualizations could demonstrate how a large and heavy log could be carefully balanced from an off-center fulcrum point. Holographic interactive simulations attempted to describe the intended awe of a precariously balanced tree while the fabrication process remained open to interpretation. Future work includes testing a variety of visualization techniques that reveal invisible factors, i.e. log density and moisture content, and implying weight and balance within interactive holograms.

Taking a more granular approach, the "texture" project drew inspiration from *naguri*; timber processing techniques that leave behind unique markings along the surface of the log. Mesh modeling and calibrated holographic visualizations were initially intended to guide the same traditional hand tools that are historically associated with *naguri*, but are equally useful to carving processes using power tools. Color gradients applied to the 3D scan data converted into customized

meshes dictate the intention to vary the depth of texture—a direct result of tool use that can only be translated by experienced hands. The speed, pressure, and strike angle of chisels, axes, and electric grinders alike are controlled by the interpretation and skill of the craftsman. Future work includes material prototyping, customizable holographic user interfaces, and collaborative hybrid sculpting.

Acknowledging calibration as an important part of MR fabrication, the "weave/thatch" project investigates the combination of the two. The study here is multi-faceted; first, to calibrate a bending simulation to real-world material parameters by overlaying a hologram modeled in K2Engineering/Kangaroo2 over an actively bent wooden lath. When aligned, the numerical inputs are less arbitrary and the simulation can more accurately reflect the bending of the material being used. Second, the project reveals the geometric rules of *kagome* basketry by projecting holographic nodes of specifically arranged triangles. By connecting the dots, a novice unaware of how the topology determines the form can learn about the mathematical concepts underlying traditional weaving techniques. Future work includes developing MR guided and novel thatching sequences for non-traditional structures.

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8 Workshop prototype installed at Yanagi Mokuzai, Hida, Japan and its digital twin of raw 3D scan data and refined meshes

and assembly. Sincere gratitude to the students involved in the workshop for their enthusiastic and clever work. Thanks to the Department of Architecture at the University at Buffalo, Hidakuma, FabCafe Tokyo, and FabCafe Hida for supporting this work-in-progress.

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IMAGE CREDITS

Figure 2, 3: Hidakuma 2022

Figure 4, 6, 8 (left): Chudy, Heiser, Nicpon 2022

Figure 5, 7: Clay, Marsh, Mohammadyar 2022

All other drawings and images by the authors.

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